

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EE)

Dec 2012

EE403 Power Electronics & Drives - II

Date: 05.12.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Draw and Explain principle of Integral cycle control and find out equation of output rms voltage. [07]

Q - 2 (a) Explain Stator Voltage and frequency control (V/f) for 3-phase Induction Motor with its characteristics. [07]

(b) Write down advantages and disadvantages of AC drives with respect to DC drives. [07]
And state various methods of speed control of an Induction motor.

OR

Q - 2 (a) Draw and explain two stage sequence controlled AC voltage controller with R load. [07]
And find out equation for rms output voltage and input pf.

(b) Draw and explain speed-torque characteristics and power-speed characteristics of a 3-phase Induction motor with variable voltage & variable frequency power supply. [07]

Q - 3 (a) A single-phase full-wave ac voltage controller feeds a load of $R=20\Omega$ with an input voltage of 230 V, 50 Hz. Firing angle for both the thyristor is 45° . Calculate (i) rms value of output voltage (ii) load power and input pf (iii) average and rms value of thyristor current. [07]

(b) Explain single-phase to single-phase step-up mid-point type cycloconverter with its necessary switching waveforms. [07]

OR

Q - 3 (a) Explain single-phase to single-phase step-down bridge type cycloconverter with R load. Also draw all gate pulses for all switching devices. [07]

(b) Draw and explain static scherbius drive for WRIM with neat diagram. [07]

SECTION - II

Q - 4 Describe 3-phase 180° mode Voltage source Inverter. Draw the waveforms of all gate pulses, line to line voltage & phase voltage. [07]

Q - 5 (a) Explain Harmonic reduction in the inverter output voltage by PWM method. [04]

(b) A three-phase bridge inverter delivers power to a resistive load from a 450 V dc source. For a star-connected load of $10\ \Omega$ per phase, determine for 120° mode, (i) rms value of load current (ii) rms value of thyristor current (iii) load power. [05]

(c) Draw and Explain Online UPS with its advantages and Applications. [05]

OR

Q - 5 (a) Draw and Explain 6 pulse converter with neat diagram and waveforms. [06]

(b) Explain Fly-back converter with neat diagram. [08]

Q - 6 (a) A single-phase ac switch of Fig.1 is used in between a 230 V, 50 Hz source and a load of 2 kW at pf of 0.8 lagging. Determine the voltage and current rating of (i) the thyristor and (ii) diodes of the bridge. Take a factor of safety of two. [06]

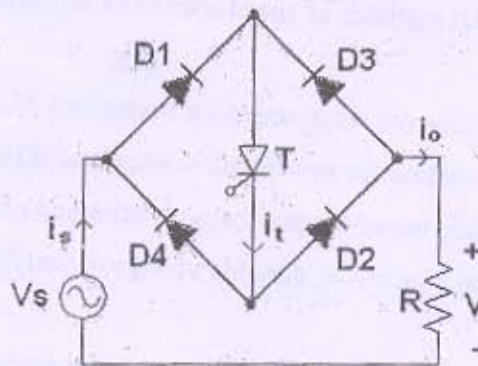


Fig.1

(b) What do you mean by Static Switches? Why they are replacing mechanical and electromechanical switches? [03]

(c) Draw and explain Full-bridge Converter with its Advantages and disadvantages. [05]

OR

Q - 6 (a) Draw and Explain 12 pulse converter with neat diagram and waveforms. [06]

(b) Explain Zero voltage Switching Resonant Converter. [08]
